

Let, $x \in A$. By Defn. of symmetric difference, $x \notin B$.

By Defn. of OR, $(x \in A \vee x \in B)$. By Defn. 1, $x \in A \cup B$.

By defn. of not belongs to, $\neg(x \in B)$ is True. By defn. of negation, $\neg(\neg(x \in B))$ is False. By Double negation Law of propositions, $x \in B$ is False.

By Defn. of AND, $(x \in A \wedge x \in B)$ is False. By Defn. 2, $x \in (A \cap B)$ is False. By Defn. of negation, $\neg(x \in A \cap B)$ is True. By Defn. of not belongs to, $x \notin (A \cap B)$.

By Defn. of AND, $x \in A \cup B \wedge x \notin A \cap B$. By Defn. 4, $x \in (A \cup B) - (A \cap B)$.

Let, $x \notin A$. By Defn. of symmetric difference, $x \in B$.

By Defn. of OR, $(x \in A \vee x \in B)$. By Defn. 1, $x \in A \cup B$.

By Defn. of not belongs to, $\neg(x \in A)$ is True. By Defn. of negation, $\neg(\neg(x \in A))$ is False. By the Double-Negation Law of Propositions, $x \in A$ is False.

By Defn. of AND, $(x \in A \wedge x \in B)$ is False. By Defn. 2, $x \in (A \cap B)$ is False.

By the defn. of negation, $\neg(x \in A \cap B)$ is True. By the defn. of not belongs to, $x \notin A \cap B$ is True. By Defn. of AND, $x \in A \cup B \wedge x \notin A \cap B$ is True. By Defn. 4, $x \in (A \cup B) - (A \cap B)$.

By Defn. 3 from Section 2.1, $A \oplus B \subseteq (A \cup B) - (A \cap B)$

Let, x be an arbitrary element of $(A \cup B) - (A \cap B)$, i.e. $x \in (A \cup B) - (A \cap B)$.

By Defn. 4, $x \in (A \cup B) \wedge x \notin (A \cap B)$. By Defn. of not belongs to, $x \in (A \cup B) \wedge \neg(x \in A \cap B)$. By Defn. 1, $(x \in A \vee x \in B) \wedge \neg(x \in (A \cap B))$.

By Defn. 2, $(x \in A \vee x \in B) \wedge \neg(x \in A \cap B)$. By 1st De Morgan's Law of propositions, $(x \in A \vee x \in B) \wedge (\neg(x \in A) \vee \neg(x \in B))$. By 1st Distributive Law of propositions, $((x \in A \vee x \in B) \wedge \neg(x \in A)) \vee ((x \in A \vee x \in B) \wedge \neg(x \in B))$.

By 2nd Commutative Law of Propositions, $(\neg(x \in A) \wedge (x \in A \vee x \in B)) \vee (\neg(x \in B) \wedge (x \in A \vee x \in B))$. By 2nd Distributive Law of Propositions, $((\neg(x \in A) \wedge x \in A) \vee (\neg(x \in A) \wedge x \in B)) \vee ((\neg(x \in B) \wedge x \in A) \vee (\neg(x \in B) \wedge x \in B))$.